Nolan Black, Ph.D. | Simulation Engineer

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Professional & Research Experience

Pasteur Labs | Scientific ML Startup (Remote) Philadelphia, PA
Simulation R&D Engineer *January 2025 – Present*

- Developed a suite of products for automated simulation, physics informed surrogates, and design optimization across CFD, structural, thermal, and multiphysics domains for aerospace, automotive, and defense industries
- Supported customer deployments and collaborations in which production engineering workflows were adapted for automation, data analysis, and surrogate development
- Designed differentiable software to perform structural optimization of pre-trained ML surrogates in aerospace applications (Python/PyTorch)
- Developed and executed simulation workflows (Ansys suite, Siemens StarCCM+, and OpenFOAM) to generate high-fidelity datasets supporting surrogate modeling for engineering design

Drexel University | GAANN Fellowship Philadelphia, PA
Graduate Research Assistant *September 2020 – January 2025*

- Developed and validated high-performance finite-element models for multiscale design optimization (C^{++}/PETSc)
- Investigated novel methods of Machine Learning (ML) for physics-informed simulation (C^{++}/Python/PyTorch)
- Integrated ML techniques to design nature-inspired metamaterials with tailorable physical properties

Lawrence-Livermore Nat. Lab. | Center for Design Optimization Livermore, CA
NSF PhD INTERN *January 2024 – May 2024*

- Executed high-performance computing workflows in a U.S. national laboratory environment
- Implemented parallel solvers for mixed finite element methods in a high-performance computing environment (C^{++})
- Developed nonlinear finite element programs for hyperelastic, hierarchical materials for applications in structural optimization

Zimmer Biomet | Zimmer Knee Creations R&D Exton, PA
Development Engineering Coop *April 2019 – April 2020*

- Developed functional medical devices using rapid prototyping to perform orthopedic procedures on the hand and wrist
- Coordinated with orthopedic surgeons to established design criteria and verify medical device designs in cadaveric lab testing
- Adapted, tested, and re-validated existing products to meet new standards of quality based on risk management

NAVSEA | Department of the Navy Philadelphia, PA
Entry-level Engineer *April 2017 – September 2018*

- Supported US Navy shipboard engineering systems through development and revision of procedures and system documentation
- Coordinated with shipboard engineering personnel to translate operational requirements into clear, auditable engineering documentation

Skills & Interests

Computational
- C++ (MPI, PETSc, mfem)
- Python (PyTorch, JAX, SciPy, NumPy)
- MATLAB

CAE Tools
- CFD: Ansys Fluent, OpenFOAM, StarCCM+
- Solid Mechanics: mfem, Ansys Mechanical
- CAD: Solidworks, Gmsh, CREO, Paraview

Education

Ph.D., Mechanical Engineering / Computer Science
Drexel University, Philadelphia, PA *September 2020 – December 2024*
Thesis: Multiscale Structural Optimization through Second-order Homogenization and Machine Learning

B.S./M.S., Mechanical Engineering
Drexel University, Philadelphia, PA *September 2016 – June 2020*